A Financial Services Perspective on Managing Scholarly Publishing Integrity

Orcid Events Webinar

October 2024

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Infolinx seeks to bring best-in-class risk and fraud management techniques developed by the financial services industry over sixty years to scholarly publishing

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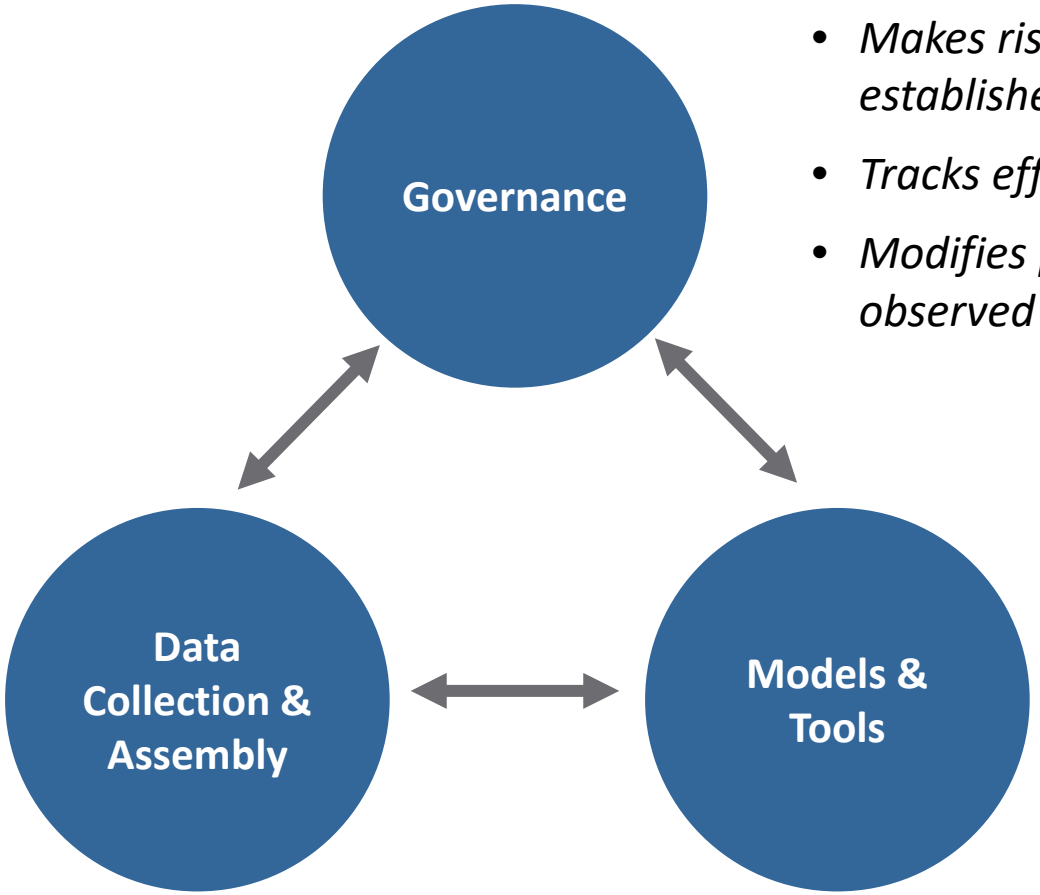
Our experience in finance indicates Integrity Management success depends on building three interrelated capabilities

- *Makes risk-reward trade-offs and establishes risk policy*
- *Tracks effectiveness of policy*
- *Modifies policy, as needed, based on observed changes and opportunities*



- *Connects the dots among a wide range of data elements*
- *Provides single source of curated data*
- *Used for tracking (against expectation), reporting and modeling*

- *Models and tools designed for specific purposes*
- *Considers all known information* ***simultaneously***

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Our approach is guided by a few overarching principles

- Focus on the actors (and connections among actors), not just symptoms
- Consider multiple factors simultaneously – a multivariate approach
- Leverage the dimension of time, and the value of time-ordering
- Develop tools that predict, rank-order and quantify the likelihood of future outcomes

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Existing tools to manage research integrity fall into two broad categories: paper content and author networks. Infolinx focuses on the latter

Integrity Management Tools	
Paper Content	**Author Network**
• Plagiarism	• Paper Mill Detectors
• Altered Graphics	• **Infolinx Paper Reliability Score**
• Tortured Language	• **Infolinx Author Network Score**
• Unusual Data Properties	

Authors are the actors; paper content symptoms of their behavior

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Our goal: leveraging author-network approaches, we seek to answer a few key questions about a paper (or submission)

- What is the likelihood an individual author would produce a particular paper?
- What is the likelihood a group of co-authors would collaborate to produce a particular paper?
- What is the likelihood the paper is legitimate?

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Our underlying premise is that deep connections among widely varied data over-time generate hard to replicate patterns – author DNA

Frequent Updating

- Authors
- Co-Authors
- Citations
- References
- Institutions
- Funders
- Professional Societies
- Conferences Attended/Presenter
- Fields of Study
- Geographic Location
- …

Evaluating all data dimensions simultaneously reduces misclassification

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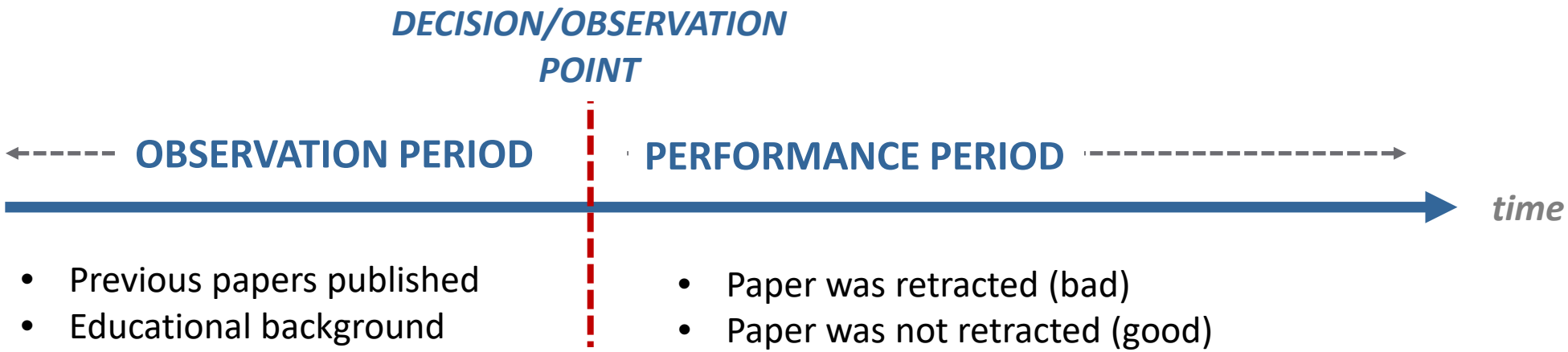
When constructing a decision tool we can consider only information known at the decision point; we evaluate an outcome over a performance period …



- Previous papers published
- Educational background
- Co-authors and background
- Employment history
- Geography
- Affiliated institutions
- Previous grants
- Co-author's papers
- Citations
- References (previous papers)
- Awards
- Comments on previous work

- Paper was retracted (bad)
- Paper was not retracted (good)

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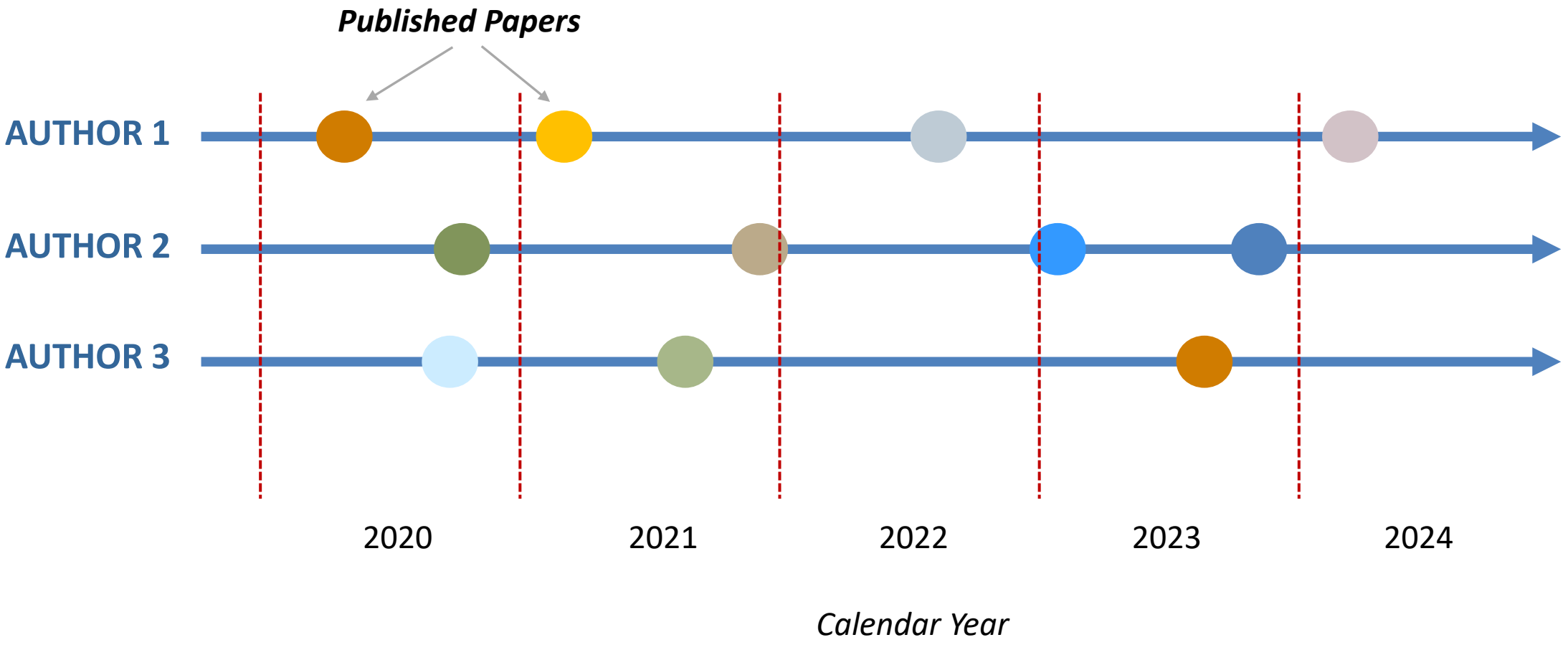
… but the existing historical record is based not on time from a decision point, but on calendar time



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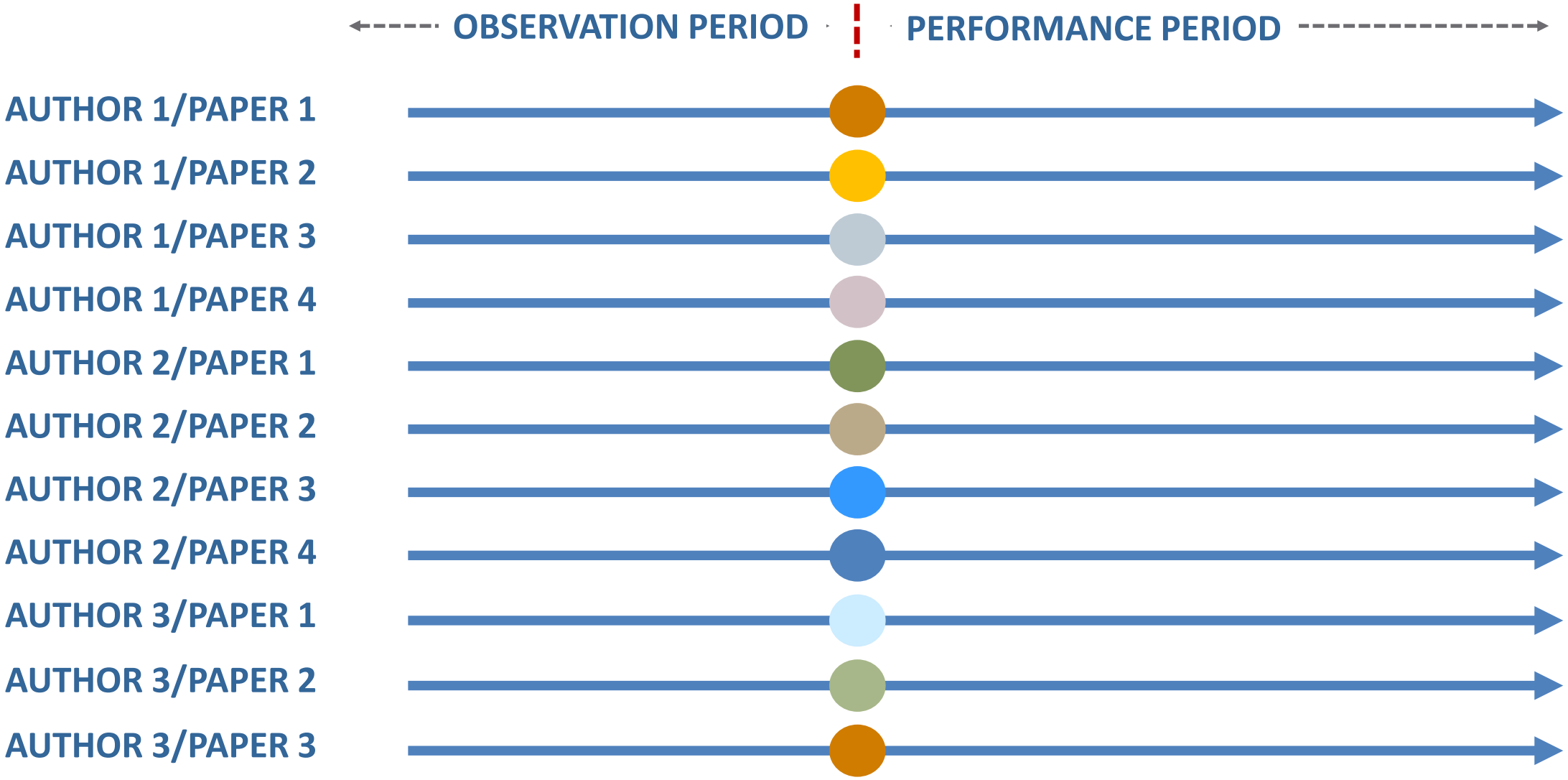
The data needs to be transformed from a calendar-oriented time framework to a paper-oriented time framework



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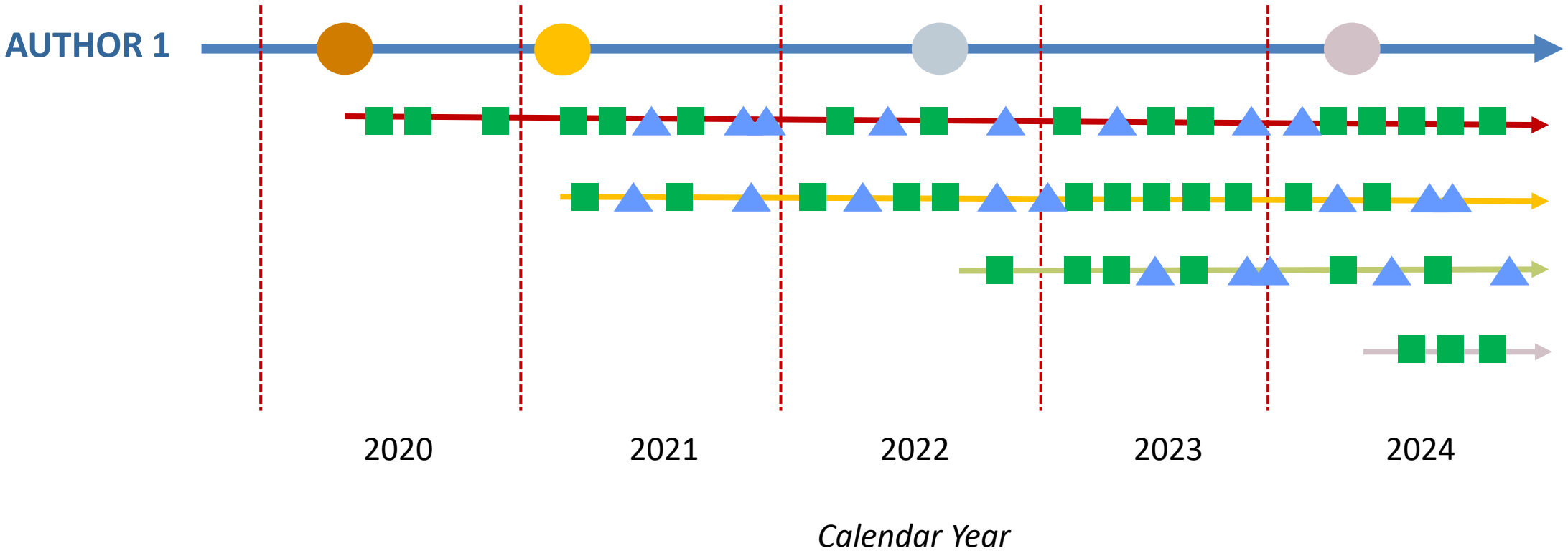
And all "events" of an author or co-author that occur after the decision point – new papers, references or comments …



Paper references

Reader comments

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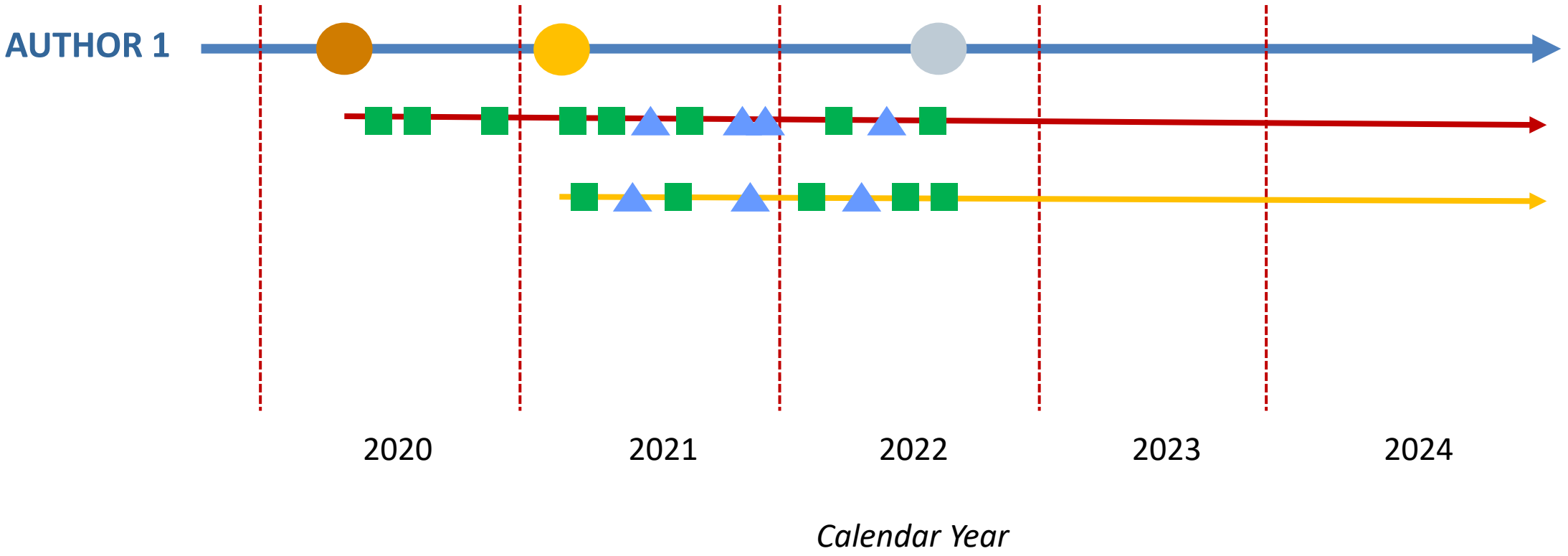

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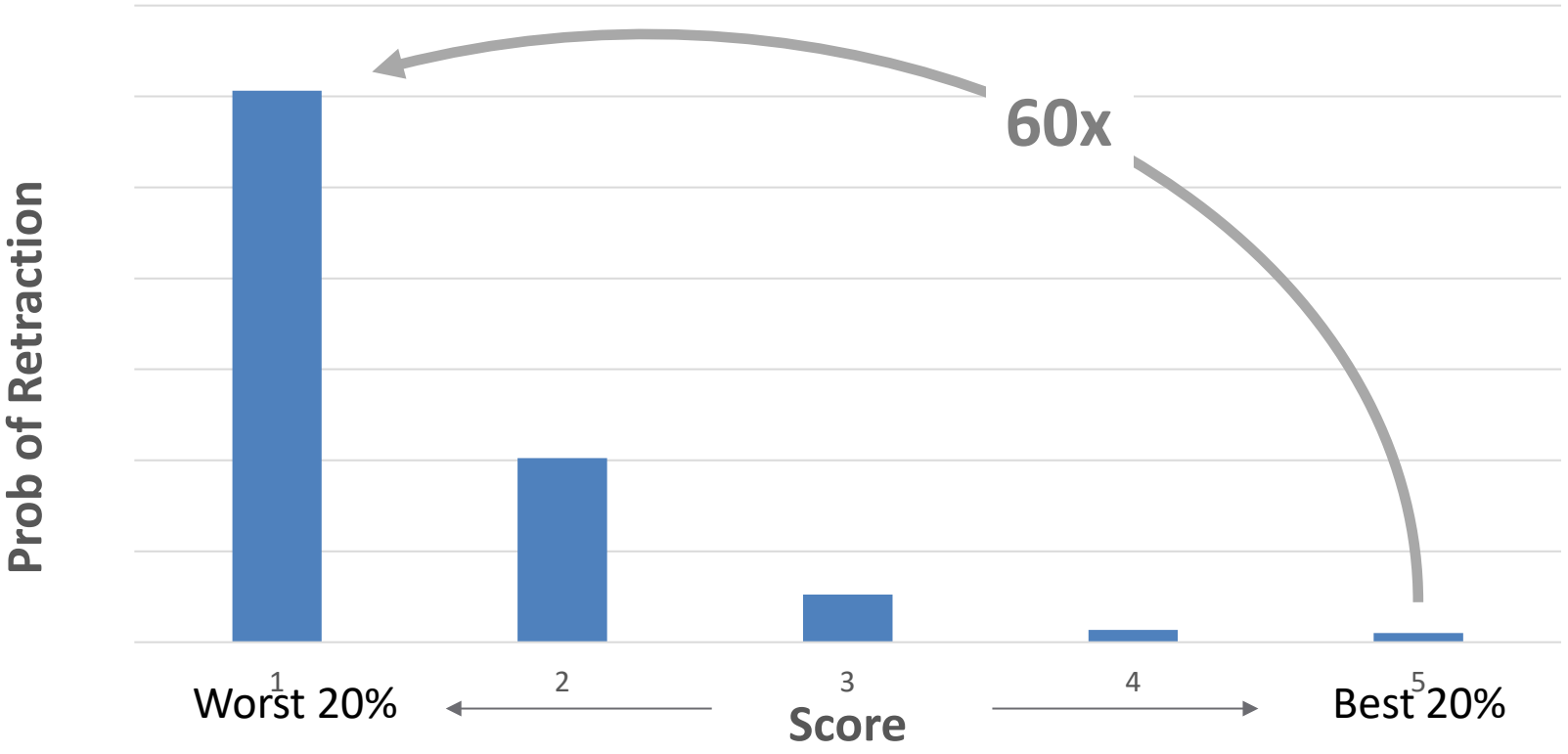
Demonstrating our approach, we built a powerful model to predict the likelihood a published paper will be retracted within ten months

The predictive model leverages six characteristics known at time of publication (our observation point) to produce a "score" for each published paper. The score's characteristics are evaluated *simultaneously* to maximize their information value and rank order the likelihood a paper will be retracted in the near-term. The worst scoring 20% of papers are 60 times more likely to be retracted than the best scoring 20%.



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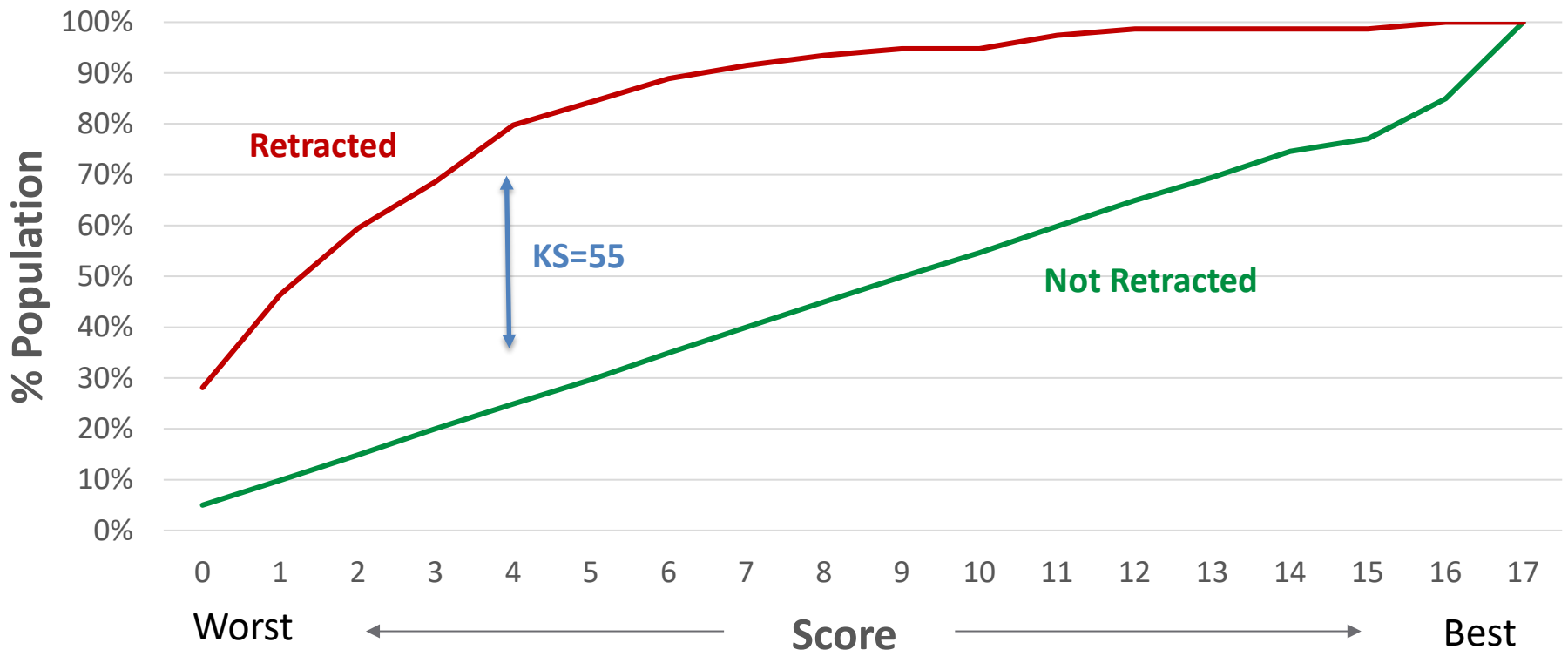
Using the score, a publisher could focus evaluative efforts at more target-rich groups of potentially problematic papers

At the point of maximum separation, denoted by a metric called KS, a publisher could capture 80% of papers that will be retracted in about a quarter of their published papers



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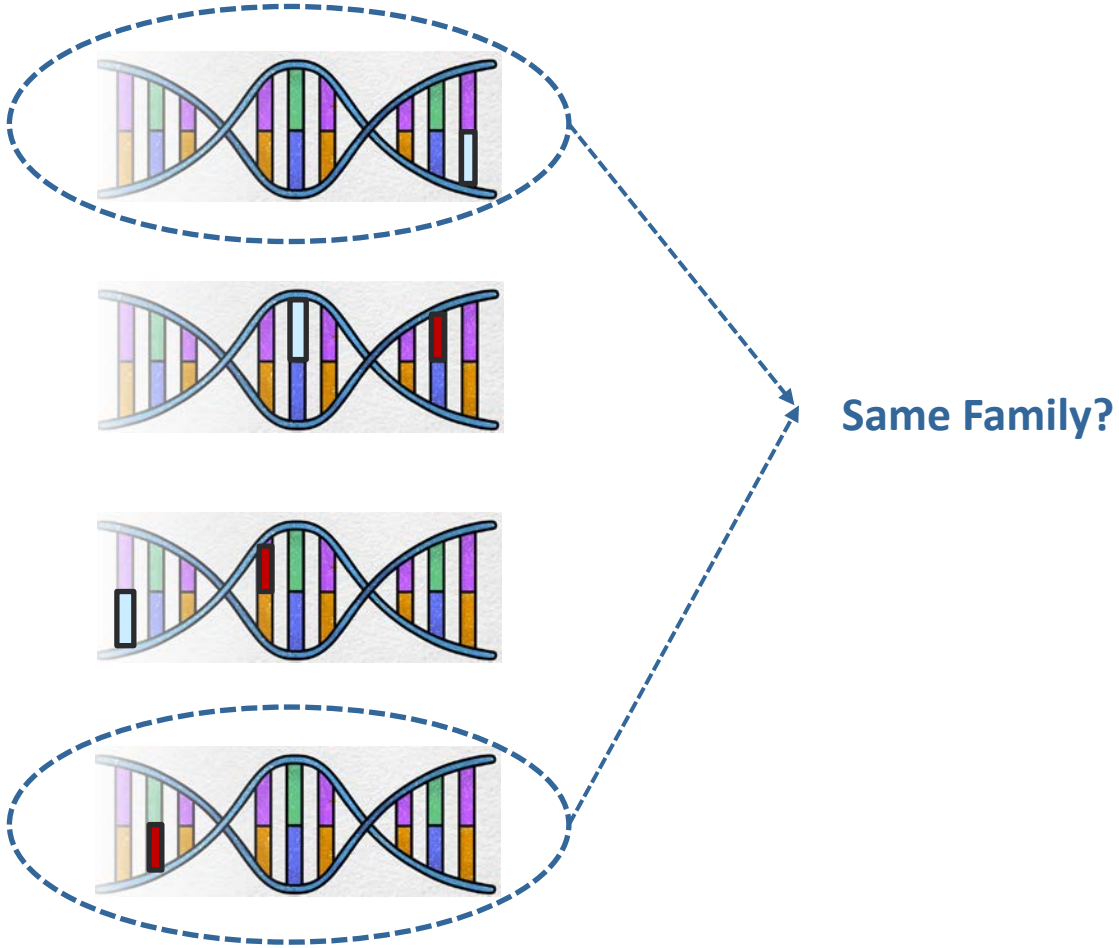
Our latest effort is to construct Author DNA to assess if a new work is consistent with an author's DNA, and if a set of authors are from the same "family"

Author DNA constructed across time from:

- **Semantic scholar (authors, papers, publication type, citations, references**
- **ISSN (journal data)**
- **Crossref (retractions, ror data, authors*, papers*)**
- **Publishers (custom built)**
- **Orcid* (author provided/verified data)**
- **Openalex* (more paper data)**

** In development*



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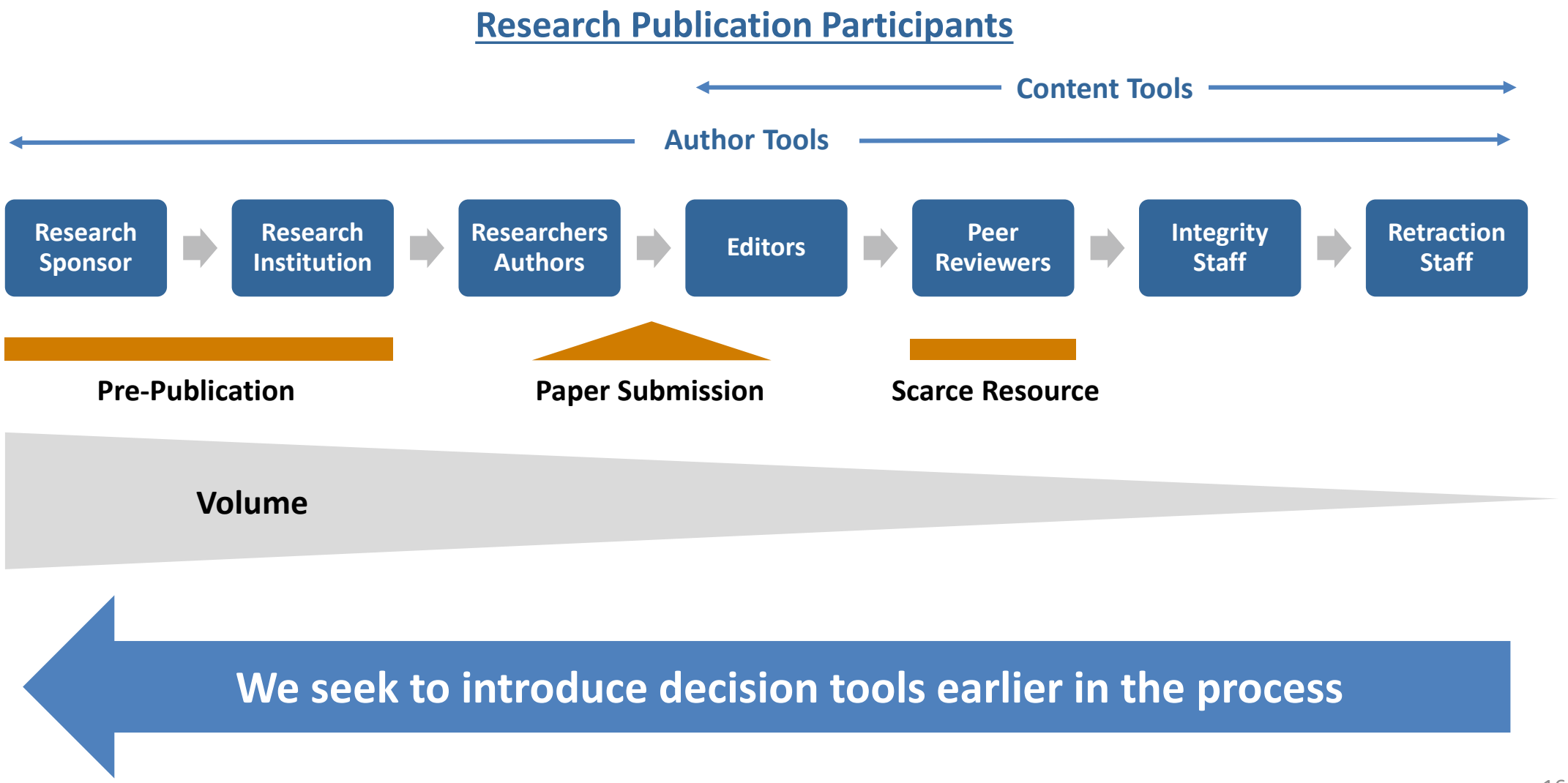
Ultimately, we believe author/scholar-centric tools can be deployed earlier in the research/publication process, creating less disruption at lower cost



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What we're working on and where we see the puck going

- Enhancing our database to be able to reconstruct history to what data looked like at any decision point. This will allow us to create proper observation and performance periods for decision model development.

- Working with publishers, institutions and funders to build custom decision tools that capture the unique nature of each business's decisions, and incorporates their unique practices, research and publication mix

- Building scholar/author DNA and tools to evaluate new works as coming from the DNA family

- Encouraging and accelerating data sharing among industry participants. We envision the industry ultimately will evolve to a shared and distributed data model where provenance, ownership and permission are incorporated into the design and sharing among participants is determined bilaterally by participants

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